Cleaning_EDA

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Load packages and data

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(GGally)

load("datA.RData")
dfA <- datA
```

Basic Structure

```r
# Dimensions
dim(dfA)
```

```
## [1] 2000   17
```

```r
# Summary
summary(dfA)
```

```
##       id                 age          menopause           race      
##  Length:2000        Min.   :25.00   Min.   :0.0000   Min.   :1.000  
##  Class :character   1st Qu.:47.00   1st Qu.:0.0000   1st Qu.:1.000  
##  Mode  :character   Median :55.00   Median :1.0000   Median :1.000  
##                     Mean   :54.99   Mean   :0.6555   Mean   :1.697  
##                     3rd Qu.:62.40   3rd Qu.:1.0000   3rd Qu.:2.000  
##                     Max.   :85.00   Max.   :1.0000   Max.   :4.000  
##       bmi          tumor_size     tumor_grade     lymph_nodes      
##  Min.   :16.00   Min.   :0.300   Min.   :1.000   Min.   : 0.000  
##  1st Qu.:23.00   1st Qu.:2.100   1st Qu.:2.000   1st Qu.: 0.750  
##  Median :26.05   Median :3.100   Median :2.000   Median : 2.000  
##  Mean   :26.06   Mean   :3.384   Mean   :2.182   Mean   : 2.623  
##  3rd Qu.:29.10   3rd Qu.:4.800   3rd Qu.:3.000   3rd Qu.: 4.000  
##  Max.   :42.60   Max.   :8.300   Max.   :3.000   Max.   :16.000  
##        ER               PR              HER2            Ki67      
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   : 1.00  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:13.50  
##  Median :1.0000   Median :0.0000   Median :0.000   Median :21.60  
##  Mean   :0.6405   Mean   :0.4805   Mean   :0.222   Mean   :21.85  
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:29.90  
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.000   Max.   :62.30  
##      stage         surgery          chemo        hormonal_therapy
##  Min.   :1.00   Min.   :0.000   Min.   :0.0000   Min.   :0.000  
##  1st Qu.:1.00   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000  
##  Median :2.00   Median :1.000   Median :1.0000   Median :0.000  
##  Mean   :2.05   Mean   :0.507   Mean   :0.5525   Mean   :0.467  
##  3rd Qu.:3.00   3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.000  
##  Max.   :3.00   Max.   :1.000   Max.   :1.0000   Max.   :1.000  
##     log_PFS
##  Min.   :-0.5345
##  1st Qu.: 1.6724
##  Median : 2.1911
##  Mean   : 2.1957
##  3rd Qu.: 2.7125
##  Max.   : 5.1013
```

Convert variables to factors

```r
dfA <- dfA |>
  mutate(
    # Nominal variables
    menopause = factor(menopause, levels = c(0,1), labels = c("Pre","Post")),
    race = factor(race, levels = c(1,2,3,4),
                  labels = c("White","Black","Asian","Hispanic")),
    surgery = factor(surgery, levels = c(0,1),
                     labels = c("BCS","Mastectomy")),
    chemo = factor(chemo, levels = c(0,1),
```

```r
                    labels = c("No","Yes")),
    hormonal_therapy = factor(hormonal_therapy, levels = c(0,1),
                              labels = c("No","Yes")),
    ER = factor(ER, levels = c(0,1), labels = c("Negative","Positive")),
    PR = factor(PR, levels = c(0,1), labels = c("Negative","Positive")),
    HER2 = factor(HER2, levels = c(0,1), labels = c("Negative","Positive")),

    # Ordinal variables
    tumor_grade = factor(tumor_grade, levels = c(1,2,3),
                         labels = c("Well","Moderate","Poor"),
                         ordered = TRUE),

    stage = factor(stage, levels = c(1,2,3),
                   labels = c("Stage I","Stage II","Stage III"),
                   ordered = TRUE)
  )
```

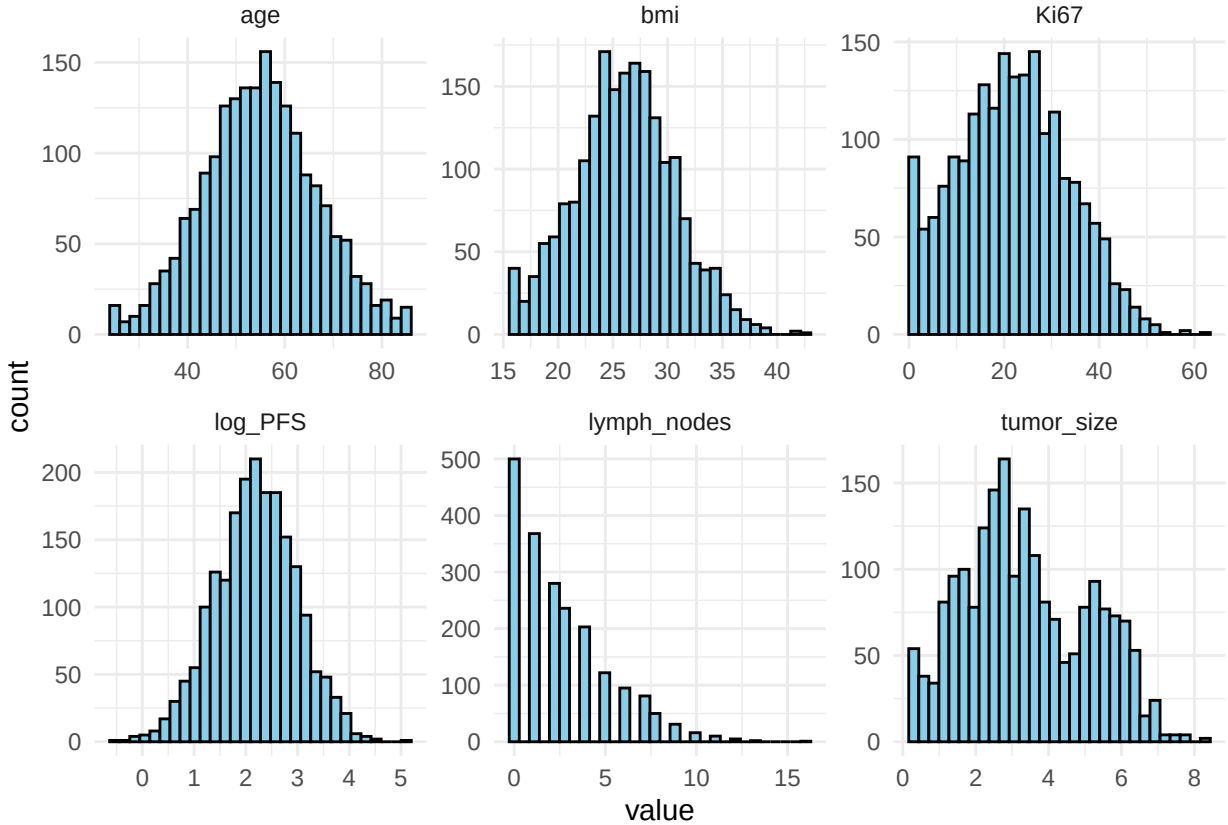
Missing data check

```r
# Count
colSums(is.na(dfA))
```

```
##               id              age        menopause             race
##                0                0                0                0
##              bmi       tumor_size      tumor_grade      lymph_nodes
##                0                0                0                0
##               ER               PR             HER2             Ki67
##                0                0                0                0
##            stage          surgery            chemo hormonal_therapy
##                0                0                0                0
##          log_PFS
##                0
```

Distribution of continuous variables

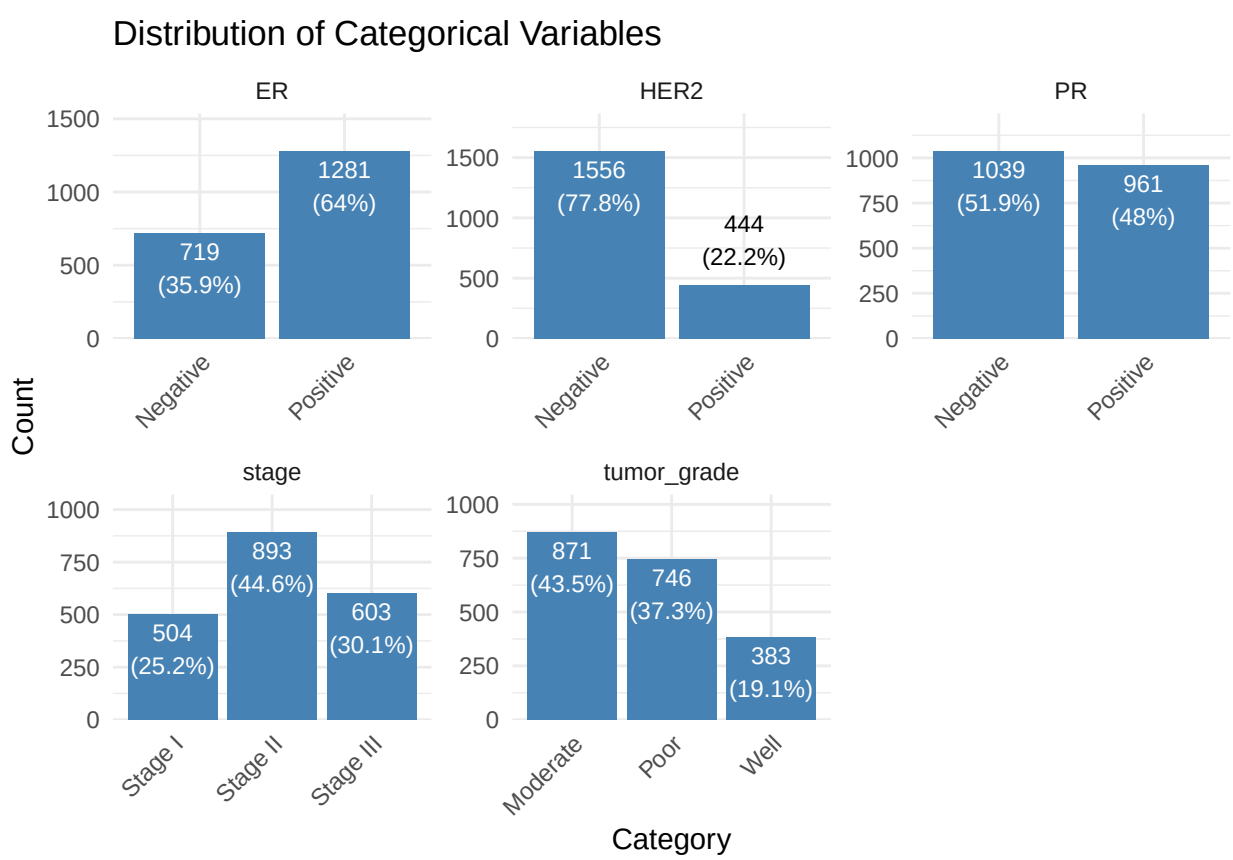
```r
# Histograms
dfA |>
  select(age, bmi, tumor_size, lymph_nodes, Ki67, log_PFS) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  facet_wrap(~name, scales = "free") +
  theme_minimal()
```



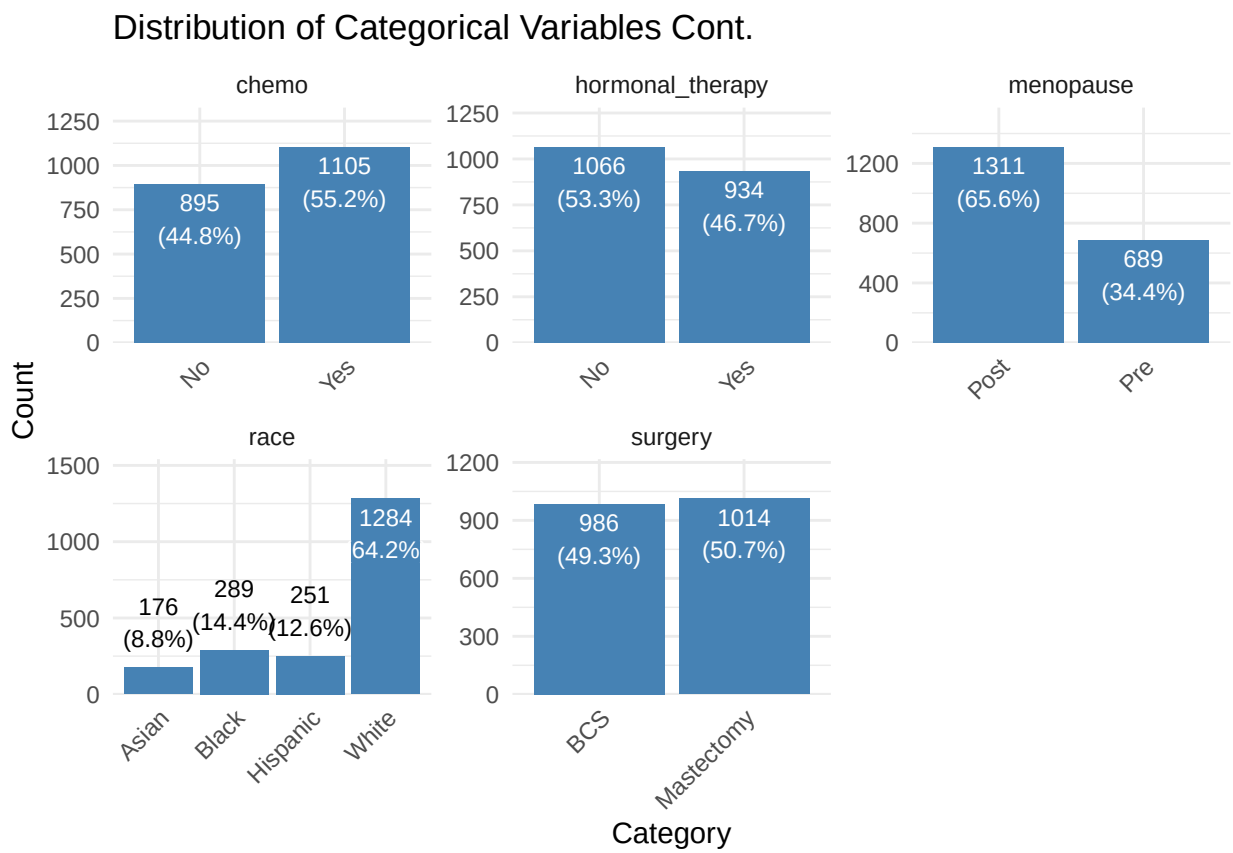
Frequency distribution of categorical variables

```
plot_categorical <- function(data, vars, title_text) {
  plot_data <- data |>
    select(all_of(vars)) |>
    mutate(across(everything(), as.character)) |>
    pivot_longer(cols = everything()) |>
    group_by(name, value) |>
    summarise(count = n(), .groups = "drop") |>
    group_by(name) |>
    mutate(
      percent = count / sum(count),
      label = paste0(count, "\n(", round(percent * 100, 1), "%)"),
      max_count = max(count)
    )
  ggplot(plot_data, aes(x = value, y = count)) +
    geom_bar(stat = "identity", fill = "steelblue") +
    geom_text(
      data = subset(plot_data, count >= max_count * 0.4),
      aes(label = label),
      vjust = 1.2,
      color = "white",
      size = 3
    ) +
    geom_text(
      data = subset(plot_data, count < max_count * 0.4),
      aes(label = label),
```

```r
      vjust = -0.4,
      color = "black",
      size = 3
    ) +
    facet_wrap(~name, scales = "free", ncol = 3) +
    scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
    theme_minimal() +
    labs(
      x = "Category",
      y = "Count",
      title = title_text
    ) +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "none"
    )
}

vars1 <- c("stage", "tumor_grade", "ER", "PR", "HER2")
vars2 <- c("race", "menopause", "chemo", "hormonal_therapy", "surgery")

plot_categorical(dfA, vars1, "Distribution of Categorical Variables")
```

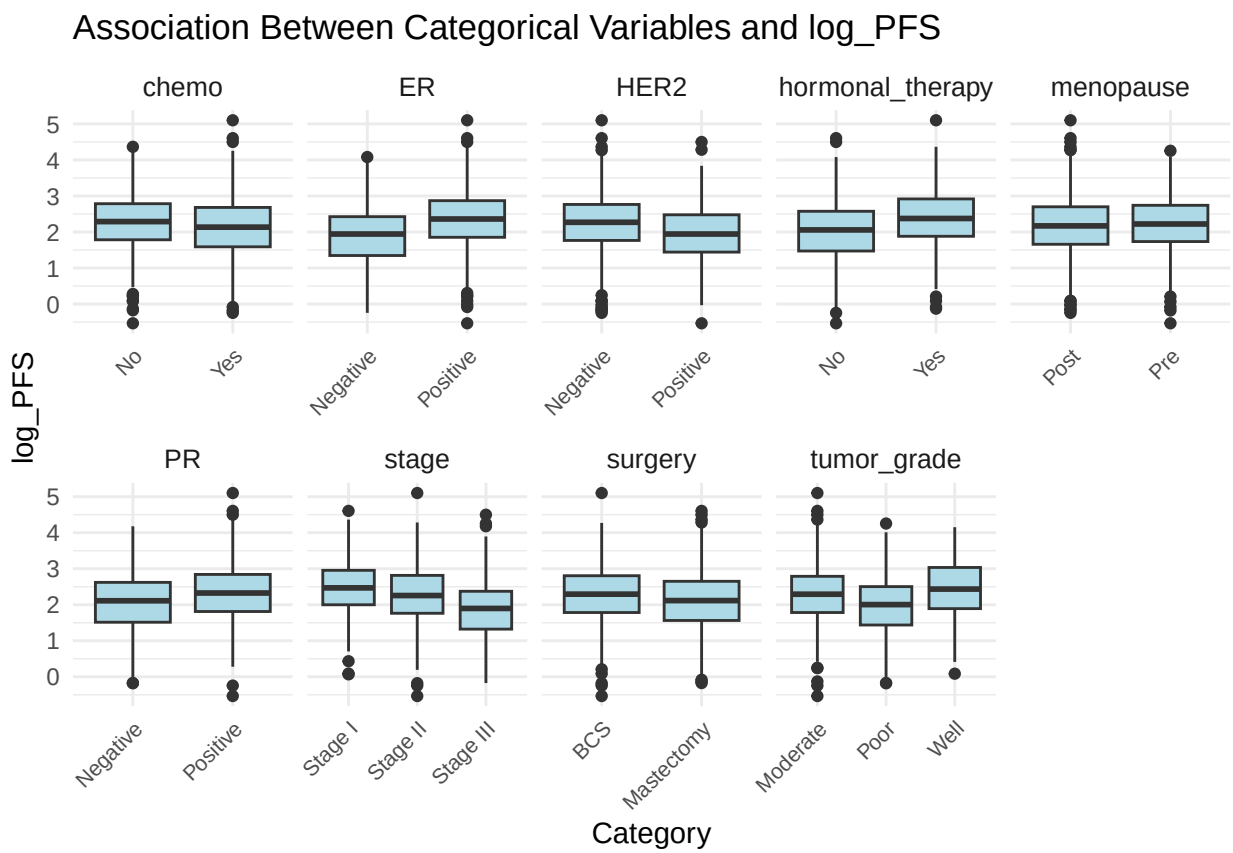



```r
plot_categorical(dfA, vars2, "Distribution of Categorical Variables Cont.")
```



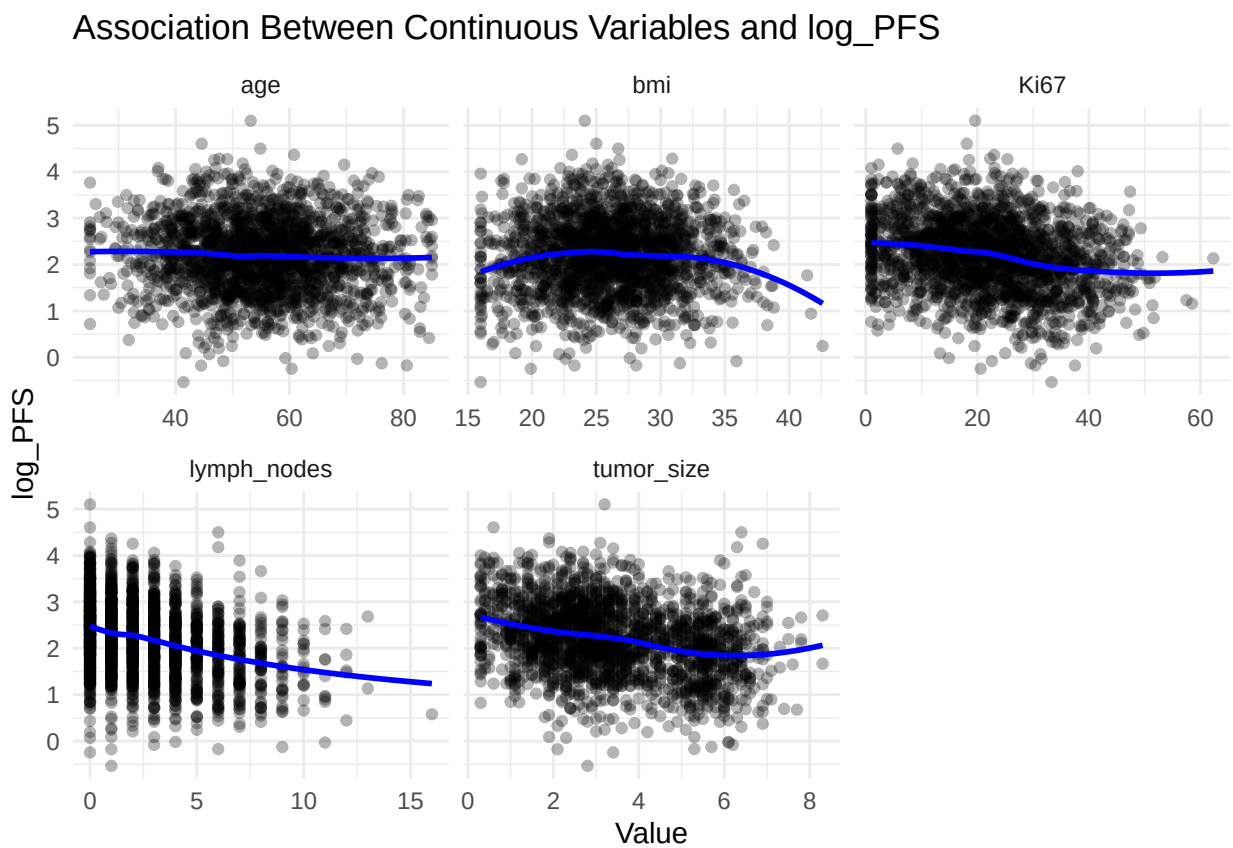
Relationship with response variable

```
# Cateforical
dfA |>
  select(log_PFS, stage, tumor_grade, ER, PR, HER2,
         chemo, hormonal_therapy, surgery, menopause) |>
  mutate(across(-log_PFS, as.character)) |>
  pivot_longer(cols = -log_PFS) |>
  ggplot(aes(x = value, y = log_PFS)) +

  geom_boxplot(fill = "lightblue") +

  facet_wrap(~name, scales = "free_x", ncol = 5) +

  theme_minimal() +
  labs(
    x = "Category",
    y = "log_PFS",
    title = "Association Between Categorical Variables and log_PFS"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(size = 10)
  )
```




```r
# Continuous
dfA |>
  select(log_PFS, age, bmi, tumor_size, lymph_nodes, Ki67) |>
  pivot_longer(cols = -log_PFS) |>
  ggplot(aes(x = value, y = log_PFS)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  facet_wrap(~name, scales = "free_x") +
  theme_minimal() +
  labs(
    x = "Value",
    y = "log_PFS",
    title = "Association Between Continuous Variables and log_PFS"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



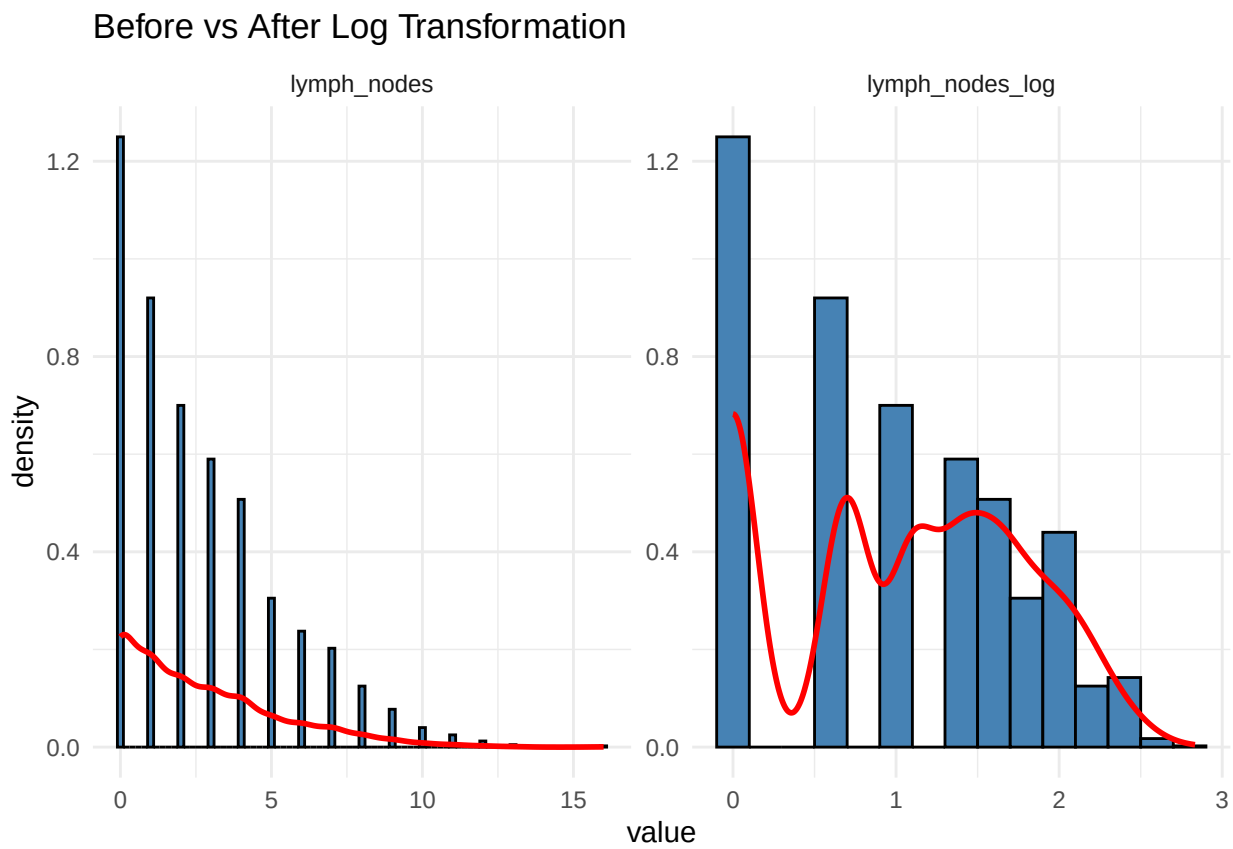
Correlation Check (Continuous)

```
cont_vars <- dfA |>
  select(age, bmi, tumor_size, lymph_nodes, Ki67, log_PFS)

cor(cont_vars, use = "complete.obs")
```

```
##                     age          bmi  tumor_size lymph_nodes        Ki67
## age          1.00000000  0.080773137  0.01855446  0.05970366 -0.04257004
## bmi          0.08077314  1.000000000  0.03323924  0.03105259 -0.03638665
## tumor_size   0.01855446  0.033239238  1.00000000  0.70632086  0.15023485
## lymph_nodes  0.05970366  0.031052587  0.70632086  1.00000000  0.14087906
## Ki67        -0.04257004 -0.036386647  0.15023485  0.14087906  1.00000000
## log_PFS     -0.05681824  0.004781066 -0.29273952 -0.32016753 -0.24940319
##                  log_PFS
## age         -0.056818244
## bmi          0.004781066
## tumor_size  -0.292739518
## lymph_nodes -0.320167531
## Ki67        -0.249403193
## log_PFS      1.000000000
```

Handle Skewness

```r
# Log transform lymph_nodes
dfA <- dfA |>
  mutate(
    lymph_nodes_log = log(lymph_nodes + 1)
  )
dfA |>
  select(lymph_nodes, lymph_nodes_log) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(aes(y = after_stat(density)),
                 binwidth = 0.2,
                 fill = "steelblue",
                 color = "black") +
  geom_density(color = "red", linewidth = 1) +
  facet_wrap(~name, scales = "free") +
  theme_minimal() +
  labs(title = "Before vs After Log Transformation")
```



Train Test Split

```
set.seed(1)

train_index <- createDataPartition(dfA$log_PFS, p = 0.7, list = FALSE)
train_data <- dfA[train_index, ]
test_data  <- dfA[-train_index, ]
```

SAVE

```
save(train_data, test_data, file = "clean_datA.RData")
```